

MARKER SELECTION

Learned gene-gene graph



$\hat{G}$ — low-rank, end-to-end

smooth x (denoise)

Expression



$x \in \mathbb{R}^N$
(N genes)

Smoothed expression

$$\tilde{x} = \hat{G} x$$

Q-Former marker router



M queries

Marker tokens



temp-annealed softmax (soft $\rightarrow$ peaked)
 $M \ll N$ (interpretable)

BIOLOGY-INFORMED RECURSIVE

ROUTING & CLASSIFICATION

weight-shared $\times K$

Shared transformer block

f_θ

ONE block

MoR depth router



primed by graph

$$+ \beta_t \cdot \pi_m$$

Mean-pool

over markers

Linear

classifier

Cell type /

phenotype

Gene-graph ablation — single-cell macro-F1



4 $\times$ fewer params
@ $K = 4$

$\sim 38\%$ fewer FLOPs
matched acc.